

What Actually Affects a Pendulum's Period?

PhET Pendulum Lab · PhET

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/pendulum-lab>
Curriculum links: energy transfer · motion · energy

Learning intentions

- I can identify which variables affect a pendulum's period and which do not.
- I can design and conduct a fair test by changing one variable at a time.
- I can explain the energy transformations occurring as a pendulum swings.

Getting started

Open PhET Pendulum Lab and set up a single pendulum. Use the stopwatch or period timer to measure how long it takes to complete a set number of swings. Change only one variable at a time — length, mass, gravity or starting amplitude — to keep your test fair.

Tasks

1. Predict which of these variables you think will affect the pendulum's period: length, mass, gravity, starting amplitude. Write down your predictions before testing.

2. Keeping mass and gravity constant, test four different pendulum lengths and record the period for each.

Length (m)	Period (s)

3. Using your table, describe the relationship between length and period. Does doubling the length double the period? Use your data to check.

Working:

4. Keeping length and gravity constant, test at least two different masses. Does mass affect the period?

5. Change the planet/gravity setting (e.g. from Earth to the Moon) while keeping length and mass constant. What happens to the period?

6. Explain, using energy transformation language (kinetic and gravitational potential energy), what is happening to the pendulum's energy as it swings from one side to the other.

7. A grandfather clock uses a pendulum to keep time. Explain why the clockmaker needs to know precisely how length affects period, but doesn't need to worry much about the mass of the pendulum bob.

8. The period of a pendulum can be estimated using $T = 2\pi\sqrt{L/g}$. Using $g = 9.8 \text{ m/s}^2$ and a length of 1 m, calculate the expected period, then compare it to what you measured in the simulation for a similar length.

Working:

Extension (optional)

Turn on friction/air resistance (if the simulation allows it) and observe what happens to the pendulum's amplitude over time. Explain, using energy transfer and transformation, where the pendulum's energy is going.
